ECON 2050 – Mathematical Economics

Basic Real Analysis II

Functions of several variables (SB Chapter 13.1)

Now we turn our attention to "functions of several variables" — functions

$$f : D \to \mathbb{R}^m$$

whose domain is some $D \subseteq \mathbb{R}^k$. Often, f is real-valued (i.e. $m = 1$). Utility, demand, production, cost and profit functions are all examples (for concrete examples, see SB pp. 274-276).

Some authors just write

$$f(x) = x^2 \quad \text{instead of} \quad f : \mathbb{R} \to \mathbb{R}, x \mapsto x^2 \quad \text{etc.},$$

and take D as the largest set of points for which f is well-defined.

Examples:

- $f(x_1, x_2) = \frac{1}{x_1 - x_2}$ has the domain $D = \{(x_1, x_2) \in \mathbb{R}^2 : x_1 \neq x_2\}$.
- $f(x_1, x_2) = \frac{1}{x_1}$ has the domain $D = \{(x_1, x_2) \in \mathbb{R}^2 : x_1 \neq 0\}$.
- $f(x_1, x_2) = \ln(x_1 + x_2)$ has the domain $D = \{(x_1, x_2) \in \mathbb{R}^2 : x_1 + x_2 > 0\}$.
- $f(x) = \sqrt{x}$ has the domain $D = \{x \in \mathbb{R} : x \geq 0\}$.

Graphs and level curves of functions (SB Chapter 13.2)

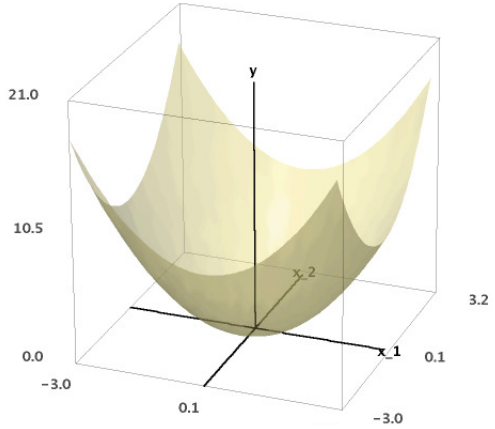
Graphing a function is often very useful if $D \subseteq \mathbb{R}^2$.

- 3-dimensional plot: graph of f
  Recall: **graph** of $f = \{(\boldsymbol{x}, y) \in D \times \mathbb{R} \subseteq \mathbb{R}^3 : y = f(\boldsymbol{x})\}$
- 2-dimensional plot: contour plot of f
  The **level curve** (or **contour line**) of level c of $f : D \subseteq \mathbb{R}^2 \to \mathbb{R}$ is the set of points

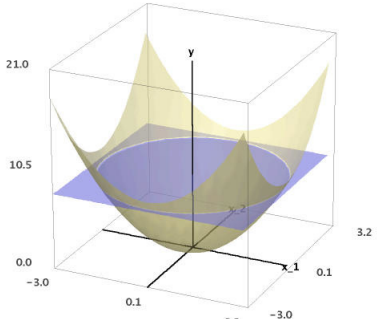
$$\{\boldsymbol{x} \in D : f(\boldsymbol{x}) = c\}.$$

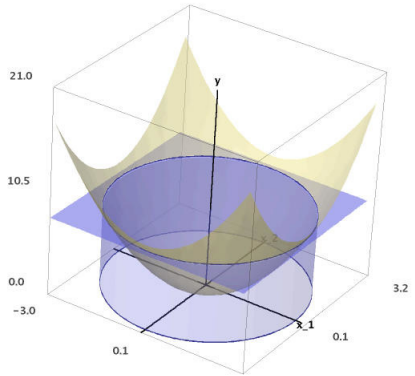
  A contour plot usually shows the level curves for several c.

Example. Below the graph of the function $f : \mathbb{R}^2 \to \mathbb{R}, (x_1, x_2) \mapsto x_1^2 + x_2^2$.

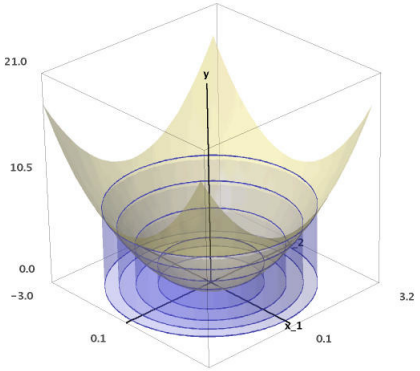


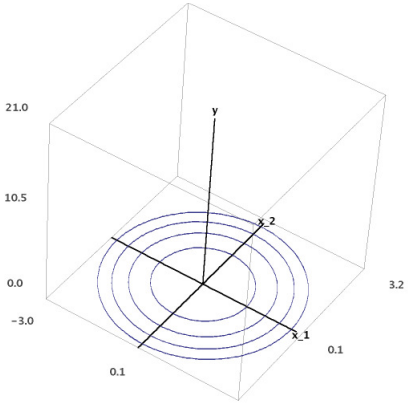
The level curve of level 8 of f is the result of first intersecting the graph of f with the horizontal plane $y = 8$ and projecting the intersection to the x_1-x_2-plane:



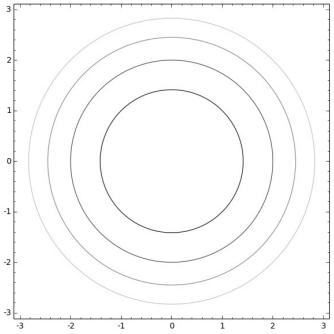


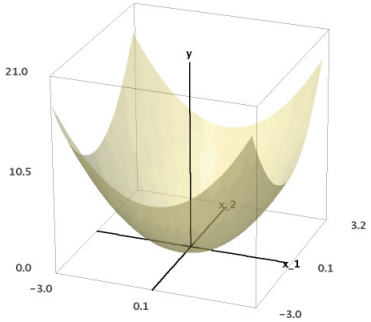
We obtain a contour plot by drawing several level curves of f.





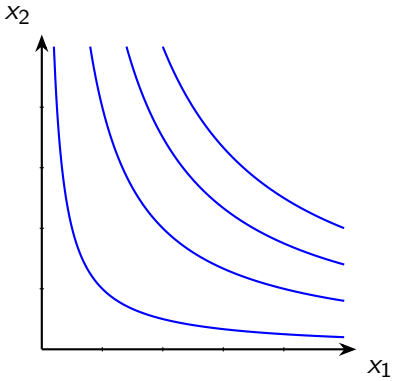
Below a contour plot and, again, the graph of f.





If you have taken intermediate microeconomics, you have already encountered level curves:

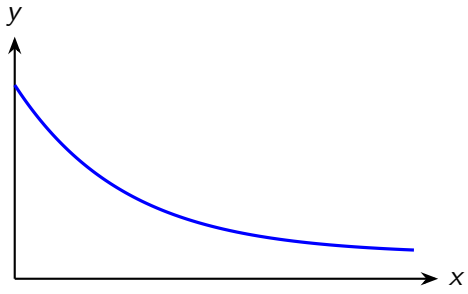
- Level curve of a utility function $u : \mathbb{R}^2_+ \to \mathbb{R} =$ indifference curve .
  Example. Cobb-Douglas utility function $u : \mathbb{R}^2_+ \to \mathbb{R}, (x_1, x_2) \mapsto x_1^{0.5} x_2^{0.5}$



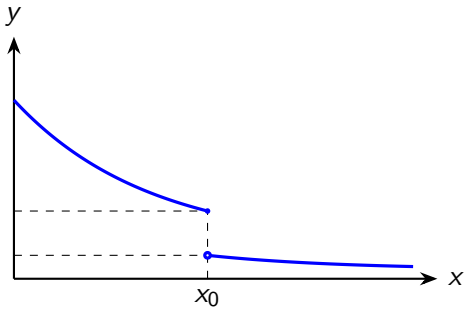
- Level curve of a production function $F : \mathbb{R}^2_+ \to \mathbb{R} =$ isoquant .

Continuity (SB Chapter 13.4)

Intuitively, a function of one variable is continuous if its graph has no breaks.



A continuous function



A discontinuous function

Formally and more generally,

Definition

Let $D \subseteq \mathbb{R}^k$. A function $f : D \to \mathbb{R}^m$ is

- **continuous at** $\boldsymbol{x} = (x_1, x_2, ..., x_k) \in D$ if for every sequence $(\boldsymbol{x}_n)$ in D that converges to $\boldsymbol{x}$, the sequence $(f(\boldsymbol{x}_n))$ in $\mathbb{R}^m$ converges to $f(\boldsymbol{x})$.
- **discontinuous at** $\boldsymbol{x}$ otherwise.
- **continuous** if it is continuous at $\boldsymbol{x}$ for all $\boldsymbol{x} \in D$.

Why do we care?

For example:

- Consumer theory predicts that the consumer will choose a (utility) maximizing bundle. Hence it is important to know that there actually is a maximizing bundle. Otherwise, what does the theory tell us?
- The following theorem guarantees that every continuous utility function has a maximum in a budget set. Hence with continuity, we are fine!
  (Recall from the discussion in the previous lecture that budget sets are compact sets.)

Theorem (Extreme Value Theorem)

Let $D \subseteq \mathbb{R}^k$ be compact and $f : D \to \mathbb{R}$ be continuous. Then f has a global maximum, that is, there exists a $\boldsymbol{x}^* \in D$ such that $f(\boldsymbol{x}^*) \geq f(\boldsymbol{x})$ for all $\boldsymbol{x} \in D$.

This result will be discussed in detail when we discuss optimization.

Proving continuity of a function at $\boldsymbol{x}$ directly may not be too straightforward as it requires checking if $f(\boldsymbol{x}_n) \to f(\boldsymbol{x})$ for *every* $(\boldsymbol{x}_n)$ that converges to $\boldsymbol{x}$.

However, if we find a single $\boldsymbol{x}_n \to \boldsymbol{x}$ such that $f(\boldsymbol{x}_n)$ *doesn't* converge to $f(\boldsymbol{x})$, then we know that f is discontinuous at $\boldsymbol{x}$.

Example: Consider the function $f : \mathbb{R}^2 \to \mathbb{R}$ with

$$f(x_1, x_2) = \begin{cases} \frac{x_1 x_2}{x_1^2 + x_2^2} & \text{if } (x_1, x_2) \neq (0, 0) \\ 0 & \text{if } (x_1, x_2) = (0, 0) \end{cases}$$

We show that it is discontinuous at $(0, 0)$.

- The sequence $(\hat{\boldsymbol{x}}_n) = (\frac{1}{n}, 0)$ converges to $(0, 0)$. Since

$$f(\hat{\boldsymbol{x}}_n) = f\left(\frac{1}{n}, 0\right) = \frac{\frac{1}{n} \cdot 0}{\left(\frac{1}{n}\right)^2 + 0^2} = 0, \qquad \forall n,$$

  the sequence $(f(\hat{\boldsymbol{x}}_n))$ converges to 0.
- $(\tilde{\boldsymbol{x}}_n) = (\frac{1}{n}, \frac{1}{n})$ also converges to $(0, 0)$, but $f(\tilde{\boldsymbol{x}}_n) \to \frac{1}{2}$ as

$$f(\tilde{\boldsymbol{x}}_n) = f\left(\frac{1}{n}, \frac{1}{n}\right) = \frac{\frac{1}{n} \cdot \frac{1}{n}}{\left(\frac{1}{n}\right)^2 + \left(\frac{1}{n}\right)^2} = \frac{1}{2}, \qquad \forall n.$$

- Therefore, we cannot have have $f(\boldsymbol{x}_n) \to 0$ for all $(\boldsymbol{x}_n)$ such that $\boldsymbol{x}_n \to (0, 0)$.

Let $D \subseteq \mathbb{R}^k$ and $E \subseteq \mathbb{R}^k$. For $f : D \to \mathbb{R}$ and $g : E \to \mathbb{R}$, define the following functions:

- $f + g : D \cap E \to \mathbb{R}, \boldsymbol{x} \mapsto f(\boldsymbol{x}) + g(\boldsymbol{x})$
- $f - g : D \cap E \to \mathbb{R}, \boldsymbol{x} \mapsto f(\boldsymbol{x}) - g(\boldsymbol{x})$
- $f \cdot g : D \cap E \to \mathbb{R}, \boldsymbol{x} \mapsto f(\boldsymbol{x})g(\boldsymbol{x})$
- $f/g : D \cap \{\boldsymbol{x} \in E : g(\boldsymbol{x}) \neq 0\} \to \mathbb{R}, \boldsymbol{x} \mapsto \frac{f(\boldsymbol{x})}{g(\boldsymbol{x})}$

The following theorem helps to show that functions are continuous.

Theorem

If $f : D \to \mathbb{R}$ and $g : E \to \mathbb{R}$ are continuous at $\boldsymbol{x} \in D \cap E$, then $f + g$, $f - g$, $f \cdot g$, and f/g (if $g(\boldsymbol{x}) \neq 0$) are all continuous at $\boldsymbol{x}$.

Proof for $f + g$: Let $(\boldsymbol{x}_n)$ be a sequence converging to $\boldsymbol{x}$.

- Because both f and g are continuous, $f(\boldsymbol{x}_n)$ converges to $f(\boldsymbol{x})$, and $g(\boldsymbol{x}_n)$ converges to $g(\boldsymbol{x})$.
- By a theorem from the previous lecture, $f(\boldsymbol{x}_n) + g(\boldsymbol{x}_n)$ converges to $f(\boldsymbol{x}) + g(\boldsymbol{x})$.
- By definition, $f + g$ is also continuous at $\boldsymbol{x}$.

The proof for $f - g$ is analogous

Theorem

Let $D \subseteq \mathbb{R}^k$ and $f = (f_1, \ldots, f_m) : D \to \mathbb{R}^m$. Then f is continuous at $\boldsymbol{x} \in D$ if and only if for every $i = 1, \ldots, m$, f_i is continuous at $\boldsymbol{x} \in D$.

This is because a theorem from the previous lecture (Slide 23) indicates that $f(\boldsymbol{x}_n) \to f(\boldsymbol{x}^*)$ if and only if $f_i(\boldsymbol{x}_n) \to f_i(\boldsymbol{x}^*)$ for each $i = 1, \ldots, k$.

Theorem

For every $i = 1, \ldots, k$, the i-th **projection** *$pr_i : \mathbb{R}^k \to \mathbb{R}, \boldsymbol{x} \mapsto pr_i(\boldsymbol{x}) = x_i$ is a continuous function.*

This is because if $\boldsymbol{x}_n \to \boldsymbol{x}^*$, then for any $i = 1, \ldots, k$, $x_{in} \to x_i^*$, where x_{in} and x_i^* are the i-th coordinate of $\boldsymbol{x}_n$ and $\boldsymbol{x}^*$, respectively. Alternatively, one may also consider $f : \mathbb{R}^m \to \mathbb{R}^m$ given by $f(\boldsymbol{x}) = \boldsymbol{x}$, and then use the theorem above.

Let $D \subseteq \mathbb{R}^k$ and $f : D \to \mathbb{R}^m$. Let $g : E \to \mathbb{R}^l$, where $f(D) \subseteq E$. Then we define the **composition** of f with g as

$$g \circ f : D \to \mathbb{R}^l, \boldsymbol{x} \mapsto g(f(\boldsymbol{x}))$$

Theorem

Let $D \subseteq \mathbb{R}^k$ and $f : D \to \mathbb{R}^m$ be continuous at $\boldsymbol{x} \in D$. Let $g : E \to \mathbb{R}^l$ be continuous at $f(\boldsymbol{x}) \in E$, where $f(D) \subseteq E$. Then $g \circ f : D \to \mathbb{R}^l$ is continuous at $\boldsymbol{x}$.

Proof: Let $(\boldsymbol{x}_n)$ be an arbitrary sequence contained in D converging to $\boldsymbol{x}$.

- By the continuity of f at $\boldsymbol{x}$, $f(\boldsymbol{x}_n) \to f(\boldsymbol{x})$.
- By the continuity of g at $f(\boldsymbol{x})$, $g(f(\boldsymbol{x}_n)) \to g(f(\boldsymbol{x}))$.
- This implies, by definition, that $g \circ f$ is continuous at $\mathbf{x}$.

The theorems we just discussed can be very powerful in proving continuity.

Example: $f : \mathbb{R}^2 \to \mathbb{R}, \boldsymbol{x} \mapsto x_1^2 + x_2^2$ is continuous.

Proof: The trick is to observe that $f = (pr_1)^2 + (pr_2)^2$:

- Recall that $pr_i : \mathbb{R}^2 \to \mathbb{R}, \boldsymbol{x} \mapsto pr_i(\boldsymbol{x}) = x_i$
- Hence $pr_i \cdot pr_i$, or, more succinctly, $(pr_i)^2$ is the function

$$(pr_i)^2 : \mathbb{R}^2 \to \mathbb{R}, \boldsymbol{x} \mapsto x_i^2,$$

and

$$(pr_1)^2 + (pr_2)^2 : \mathbb{R}^2 \to \mathbb{R}, \boldsymbol{x} \mapsto ((pr_1)^2(\boldsymbol{x}) + (pr_2)^2(\boldsymbol{x})) = x_1^2 + x_2^2.$$

Therefore, $(pr_1)^2 + (pr_2)^2$ equals f. By the previous theorems,

- pr_i is continuous, and therefore,
- pr_i^2 is continuous, and therefore,
- $(pr_1)^2 + (pr_2)^2$ is continuous (that is, f is continuous!)

Alternatively, we can prove that $f : \mathbb{R}^2 \to \mathbb{R}, \boldsymbol{x} \mapsto x_1^2 + x_2^2$ is continuous by realizing that $f = g \circ pr_1 + g \circ pr_2$ for $g : \mathbb{R} \to \mathbb{R}, x \mapsto x^2$.

Exercises:

- Let $D = \mathbb{R}^3 \backslash \{(x, y, z) \in \mathbb{R}^3 : x + y + z = 0\}$. Then

$$\hat{f} : D \to \mathbb{R}, (x, y, z) \mapsto \frac{xyz}{x + y + z}$$

is continuous.
Proof:

$$\hat{f} = \frac{pr_1 \cdot pr_2 \cdot pr_3}{pr_1 + pr_2 + pr_3}$$

- $\tilde{f} : \mathbb{R}^2 \backslash \{(0, 0)\} \to \mathbb{R}, (x, y) \mapsto \ln(x^2 + y^2)$ is continuous.
Proof:

$$\tilde{f} = \ln \circ f,$$

where $f : \mathbb{R}^2 \backslash \{(0, 0)\} \to \mathbb{R}, \boldsymbol{x} \mapsto x_1^2 + x_2^2$, and both f (see above) and ln are continuous.

Linear functions (SB pp. 287-289)

The **inner product** (or **dot product**) on $\mathbb{R}^k$ of $\boldsymbol{x} \in \mathbb{R}^k$ and $\boldsymbol{y} \in \mathbb{R}^k$ is

$$\boldsymbol{x} \cdot \boldsymbol{y} = \sum_{i=1}^{k} x_i y_i = x_1 y_1 + x_2 y_2 + \ldots + x_k y_k.$$

A function $f : \mathbb{R}^k \to \mathbb{R}$ is **linear** if and only if there exists $\boldsymbol{a} \in \mathbb{R}^k$ such that

$$f(\boldsymbol{x}) = \boldsymbol{a} \cdot \boldsymbol{x} \ (= \sum_{i=1}^{k} a_i x_i)$$

for all $\boldsymbol{x} \in \mathbb{R}^k$.

When $k = 1$, a linear function satisfies $f(x) = ax$ for some $a \in \mathbb{R}$, where a is the *slope* (rise over run) of f.

Affine functions (SB p. 292)

A function $f : \mathbb{R}^k \to \mathbb{R}$ is **affine** if there exist $\boldsymbol{a} \in \mathbb{R}^k$ and $b \in \mathbb{R}$ such that $f(\boldsymbol{x}) = \boldsymbol{a} \cdot \boldsymbol{x} + b$ for all $\boldsymbol{x} \in \mathbb{R}^k$.

For all $\boldsymbol{x}_0, \boldsymbol{x} \in \mathbb{R}^k$ and every affine $f : \mathbb{R}^k \to \mathbb{R}, \boldsymbol{x} \mapsto \boldsymbol{a} \cdot \boldsymbol{x} + b$,

$$\begin{aligned} f(\boldsymbol{x}) - f(\boldsymbol{x}_0) &= \boldsymbol{a} \cdot \boldsymbol{x} + b - (\boldsymbol{a} \cdot \boldsymbol{x}_0 + b) \\ &= \boldsymbol{a} \cdot \boldsymbol{x} + b - \boldsymbol{a} \cdot \boldsymbol{x}_0 - b \\ &= \boldsymbol{a} \cdot \boldsymbol{x} - \boldsymbol{a} \cdot \boldsymbol{x}_0 \\ &= \boldsymbol{a} \cdot (\boldsymbol{x} - \boldsymbol{x}_0) \end{aligned}$$

When $k = 1$, an affine function satisfies $f(x) = ax + b$ for some $a, b \in \mathbb{R}$.

Letting $\mathrm{d}\boldsymbol{x} \equiv \boldsymbol{x} - \boldsymbol{x}_0$ and $\mathrm{d}y \equiv f(\boldsymbol{x}) - f(\boldsymbol{x}_0)$, the last equation becomes

$$\mathrm{d}y = \boldsymbol{a} \cdot \mathrm{d}\boldsymbol{x} \ (= \sum_{i=1}^{k} a_i \, \mathrm{d}x_i).$$

Therefore, if f is affine, the change $\mathrm{d}y$ in the value of f is a linear function of the change $\mathrm{d}\boldsymbol{x}$ in its variables.

Exercise: Suppose $f(x_1, x_2) = 0.5x_1 + 1.25x_2 + 1.75$.

How does the value of f change if we move from $\boldsymbol{x}_0 = (1, 0.5)$ by $\mathrm{d}\boldsymbol{x} = (\mathrm{d}x_1, \mathrm{d}x_2) = (1.5, 1)$ to $\boldsymbol{x} = \boldsymbol{x}_0 + \mathrm{d}\boldsymbol{x} = (2.5, 1.5)$?

It changes by

$$\mathrm{d}y = 0.5\mathrm{d}x_1 + 1.25\mathrm{d}x_2 = 0.5 \cdot 1.5 + 1.25 \cdot 1 = 2$$

Note that we can "decompose" the change to f from moving from $\boldsymbol{x}_0 = (1, 0.5)$ to $\boldsymbol{x}_0 + \mathrm{d}\boldsymbol{x} = (2.5, 1.5)$ into

- the change from moving just in the x_1 direction (by $\mathrm{d}x_1 = 2.5 - 1 = 1.5$),

$$\mathrm{d}y_1 = 0.5\mathrm{d}x_1 = 0.5 \cdot 1.5 = 0.75,$$

plus

- the change from moving just in the x_2 direction (by $\mathrm{d}x_2 = 1.5 - 0.5 = 1$),

$$\mathrm{d}y_2 = 1.25\mathrm{d}x_2 = 1.25 \cdot 1 = 1.25.$$

The total change to f is

$$\mathrm{d}y = \mathrm{d}y_1 + \mathrm{d}y_2 = 2$$

An analogous decomposition holds for changes in the value of affine functions in general:

Recall that

$$\mathrm{d}y = \boldsymbol{a} \cdot \mathrm{d}\boldsymbol{x} \; (= \sum_{i=1}^{k} a_i \, \mathrm{d}x_i)$$

if we move by $\mathrm{d}\boldsymbol{x}$.

If we just move by $\mathrm{d}x_i$ (that is, just in the x_i direction), from any $\boldsymbol{x}_0$ to

$$\boldsymbol{x}_0 + (0, \ldots, 0, \mathrm{d}x_i, 0, \ldots, 0) = (x_{0,1}, \ldots, x_{0,i} + \mathrm{d}x_i, \ldots, x_{0,k}),$$

then

$$\mathrm{d}y_i = f(\boldsymbol{x}_0 + (0, \ldots, 0, \mathrm{d}x_i, 0, \ldots, 0)) - f(\boldsymbol{x}_0) = a_i \, \mathrm{d}x_i$$

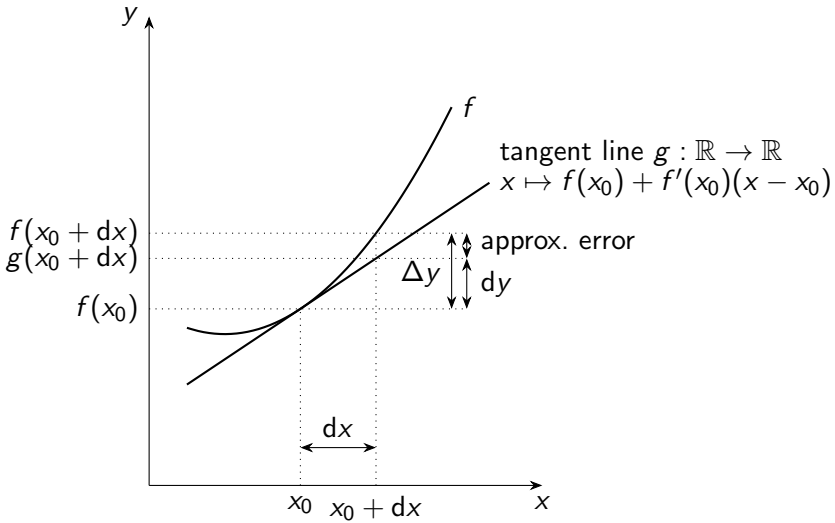
Hence we can write

$$\mathrm{d}y = \sum_{i=1}^{k} \mathrm{d}y_i$$

The change from moving from $\boldsymbol{x}_0$ to $\boldsymbol{x}_0 + \mathrm{d}\boldsymbol{x} = \boldsymbol{x}_0 + \mathrm{d}(x_1, \ldots, x_k)$ equals the sum of the changes of

- just moving in the x_1 direction, by $\mathrm{d}x_1$, and
- just moving in the x_2 direction, by $\mathrm{d}x_2$, and
- ...
- just moving in the x_k direction, by $\mathrm{d}x_k$.

Differentiability: Function of one variable (a review)



In the previous slide:

- around x_0, f is approximated by the *affine* function g tangent to f at x_0
- $f'(x_0)$ is the *slope* of the tangent line g
- $\Delta y = f(x_0 + \mathrm{d}x) - f(x_0) =$ actual change in y due to change $\mathrm{d}x$ in x
- $\mathrm{d}y = g(x_0 + \mathrm{d}x) - g(x_0) = f'(x_0)\mathrm{d}x =$ approximate change in y due to change $\mathrm{d}x$ in x; note that $\mathrm{d}y$ is a *linear* function of $\mathrm{d}x$
- if $\mathrm{d}x$ is "small,"

$$\Delta y \approx \mathrm{d}y = f'(x_0)\mathrm{d}x,$$

  and $f'(x_0)$ measures how much the change $\mathrm{d}x$ in x affects $y = f(x)$
- as $\mathrm{d}x$ approaches 0, the approximation error also goes to 0

The formal definition of differentiability is as follows:

Definition

Let $D \subseteq \mathbb{R}$ be open. The function $f : D \to \mathbb{R}$ is differentiable at $x_0 \in D$ if there exists $f'(x_0) \in \mathbb{R}$ such that

$$f'(x_0) = \lim_{n\to\infty} \frac{f(x_0 + h_n) - f(x_0)}{h_n}$$

for all sequences (h_n) with $h_n \to 0$ and $h_n \neq 0$ for all n. In this case, $f'(x_0)$ is called the **derivative** of f at x_0.

It is useful to rephrase this definition a bit:

- $f : D \to \mathbb{R}$ is differentiable at x_0 if and only if there exists $a \in \mathbb{R}$ such that

$$\lim_{n\to\infty} \frac{f(x_0 + h_n) - f(x_0) - ah_n}{h_n} = 0$$

for all sequences (h_n) with $h_n \to 0$ and $h_n \neq 0$ for all n. In this case, $a = f'(x_0)$.

- $f : D \to \mathbb{R}$ is differentiable at x_0 if and only if there exists $a \in \mathbb{R}$ such that

$$\lim_{n\to\infty} \frac{f(x_n) - f(x_0) - a(x_n - x_0)}{x_n - x_0} = 0$$

for all sequences (x_n) with $x_n \to x_0$ and $x_n \neq x_0$ for all n. In this case, $a = f'(x_0)$.

- If $f : D \to \mathbb{R}$ is differentiable at x_0, then the tangent line of f at x_0, given by

$$g : \mathbb{R} \to \mathbb{R}, x \mapsto f(x_0) + f'(x_0)(x - x_0)$$

is an affine function that approximates f around x_0 in the sense that whenever $|x - x_0|$ is sufficiently small, the "approximation error" $|f(x) - g(x)|$ is negligible.

Example: Let $f : \mathbb{R} \to \mathbb{R}$ be given by $f(x) = x^2$.

By definition, fix any $x \in \mathbb{R}$ and $h_n \to 0$ (but $h_n \neq 0$ for all n), we have

$$\frac{f(x + h_n) - f(x)}{h_n} = \frac{(x + h_n)^2 - x^2}{h_n} = \frac{x^2 + 2xh_n + h_n^2 - x^2}{h_n} = 2x + h_n$$

But then $f'(x) = \lim_{n\to\infty}(2x + h_n) = 2x$.

Example: Let $f : \mathbb{R} \to \mathbb{R}_+$ be given by $f(x) = |x|$.

This function is not differentiable at $x_0 = 0$:

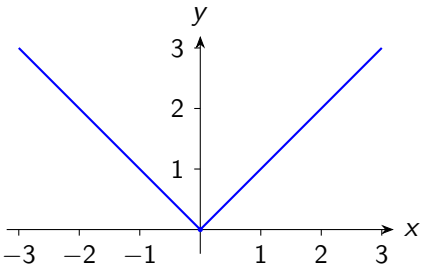
- Consider the sequence $(1/n)$,

$$\frac{f(0 + 1/n) - f(0)}{1/n} = \frac{|0 + 1/n| - |0|}{1/n} = \frac{1/n}{1/n} = 1.$$

- On the other hand, consider the sequence $(-1/n)$,

$$\frac{f(0 + (-1/n)) - f(0)}{-1/n} = \frac{|0 - 1/n| - |0|}{-1/n} = \frac{1/n}{-1/n} = -1.$$

Observe that $f(x) = |x|$ is continuous because its graph has no breaks:



In fact:

Theorem

If $D \subseteq \mathbb{R}$ is open and $f : D \to \mathbb{R}$ is differentiable at a point $x_0 \in D$, then f is continuous at x_0.